## Proposal Title:

Student Behavior Analysis: AI-Based Real-Time Student Behavior and Conflict Early-Warning System

## Project Brief:

### Real-Time Campus Security and Video Monitoring System

University security staff have to keep an eye on many CCTV cameras throughout the day. It is not always possible to watch every screen continuously, which means that the fights, suspicious movements, or entry into restricted areas may go unnoticed until someone report them and also the detect if someone tries to Bother/harass a female student. This project aims to develop a system that Can help security staff identify such situations more Quickly.

The System will use computer vision and deep learning to

analyze live camera, or CCTV streams. A pretrained YOLOv8 pose model will be used to detect people, follow their movements, and identify important body points, such as hands, arms, and shoulders. By studying this movement, the system will look for possible signs of a physical fight, including people standing too close to each other, sudden hand movements, and raised arms. it will also identify unusual crowding, entry into restricted areas, and people staying in a particular location longer than allowed.

Whenever the System detects a possible security issue, it will generate an alert with a low, medium, high severity level. The alert will include the time it occurred. Security staff will be able to view these alerts through a web dashboard and mark each one as a confirmed incident or a false alarm. This System only assist security personnel in make decisions.

The project will be developed and tested using publicly available datasets, including RWF-2000, Hockey Fights, and UCF-Crime. Its performance will be evaluated by measuring precision, recall, false-positive rate, and processing speed in frame per second.

The main goal of this project is to provide university security staff with an additional tool for monitoring campus activates, responding to possible incidents and improving safety without removing the need for human judgment.

## Technologies:

Python, OpenCV, YOLOv8, PyTorch, MediaPipe, Streamlit, SQLite

## GitHub:

https://github.com/Talha923-star/Student_Behavior_Analysis